

Figure 18.1 Different levels of organization

a multicellular organism. From lower organism to higher organism life has passed through simple to complex structural hierarchical level. The pictorial representation of biological organisation is shown in Figure 18.1.

18.2 Cells

Cell is the **structural and functional unit of life**. Cells are often called as 'building blocks of life'. The study of cells is called **cell biology**. Cells consist of cytoplasm enclosed within a membrane, which contains many biomolecules such as proteins and nucleic acids. Cells vary widely in shape and size. There is a central spherical **nucleus** and a variety of cytoplasmic living **cell organelles** like the endoplasmic reticulum, mitochondria. golgi bodies, centrioles, ribosomes, lysosomes, etc., present in an animal cell. Each cell organelle performs a specific function.

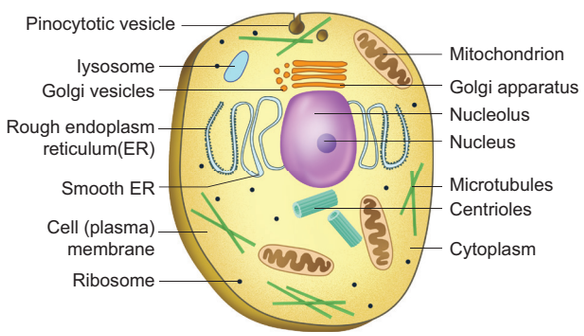
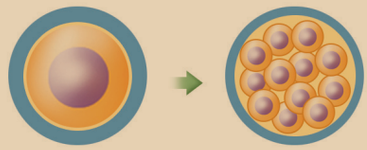


Figure 18.2 Animal cell

The size of cells varies in different animals and they which are measured in units of micron (μm). One micron is equal to 1/1000000 meter. The average cell size varies from 0.5 to 20 μm in diameter. The cells of bacteria are the smallest in size (1-2 μm). In human body, the smallest cell is red blood cells (7 μm in diameter) and the longest one is the nerve cell which reaches a length of about 90 - 100 cm. Human egg (Ovum) is 100 μm in size. Among multicellular animals, the largest cell is, egg of an ostrich. Mycoplasma with a diameter of 0.0001 mm is the smallest bacterium.

DO YOU KNOW?

Our body is developed from a single cell called zygote. The zygote undergoes continuous mitotic division and forms the foetus consisting multitude of cells of different shape, size and content. Foetal cells gradually attain change in structure and function. This process is known as cell differentiation.



18.2.1 Shape of cells

Cells are of different shapes. Normally they are correlated with their functions. Some cells are oval or round, while certain others are elongated. Some are branched like the nerve cell or a neuron. Some of our white blood cells are amoeba like with irregular boundaries.

Activity 1

Boil a hen's egg and remove the shell. What do you observe? A white material surrounds the yellow part. White material is albumin which solidifies on boiling. The yellow part is yolk. It is a part of the single cell. You can observe this single cell without any magnifying devices.

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